

Equations used in the drift & coalescent tutorials

Evolutionary Thinking, week 39

Conventions

- The notebooks simulate a **haploid** population: N individuals, each carrying one copy of the gene, so N gene copies in total. The lecture uses a **diploid** population of N individuals, i.e. $2N$ gene copies. The results are the same once you count gene copies, so every lecture formula is obtained from the notebook formula by replacing N with $2N$. The lecture form is given in grey after each result.
- There are two alleles. We call them A and a ; the notebooks call A the “derived allele” (the one that arose by mutation) and a the ancestral one. p is the frequency of A , $k = pN$ is the number of copies of A , and $q = 1 - p$.
- s is the selection coefficient, m the migration rate, n the number of gene copies in a *sample* (as opposed to N , the whole population), t time in generations.
- “The lecture” formulas marked *take as given* come from diffusion theory, which is beyond this course; everything else is derived here.

1 The Wright–Fisher model (forward in time)

1.1 What the model assumes, precisely

Generations do not overlap and the population size is constant. Every individual of generation t produces a very large number of gametes, so the **gamete pool** contains the two alleles in exactly the same proportions as generation t : a fraction p of the gametes carry A . Generation $t + 1$ is formed by drawing N gametes at random from this pool (the pool is so large that the draws do not change its composition – “sampling with replacement”). There is no selection (every gamete is equally likely to be drawn), no mutation, no population structure.

1.2 One generation is one binomial draw

Because each of the N drawn gametes carries A with probability p , independently of the others, the number of A copies in generation $t + 1$ is a binomial random variable:

$$k_{t+1} \sim \text{Binomial}(N, p), \quad p_{t+1} = \frac{k_{t+1}}{N}.$$

This is literally the line `p = np.random.binomial(N, p) / N` in the notebook: `binomial(N, p)` returns how many of N independent draws with success probability p were successes.

1.3 The expected frequency does not change; the variance is $p(1 - p)/N$

A $\text{Binomial}(N, p)$ variable has mean Np and variance $Np(1 - p)$. Dividing a random variable by the constant N divides its mean by N and its variance by N^2 , therefore

$$E[p_{t+1}] = \frac{Np}{N} = p, \quad \text{Var}(p_{t+1}) = \frac{Np(1 - p)}{N^2} = \frac{p(1 - p)}{N} \quad (\text{lecture, diploid: } p(1 - p)/2N).$$

Meaning. On average the frequency stays where it is: drift has no preferred direction. What drift does is add random scatter, and the size of that scatter shrinks in proportion to $1/N$. That is the entire content of the “size matters” slide. Over t generations the random steps accumulate, so the frequency wanders by roughly $\sqrt{tp(1 - p)/N}$ (only roughly, because $p(1 - p)$ itself changes along the way): about 0.16 after 100 generations with $N = 1000$, but 0.05 with $N = 10\,000$.

2 Fixation and loss

2.1 The probability of eventual fixation equals the current frequency

First, why does every allele end up fixed or lost? Once $p = 0$ or $p = 1$ the binomial draw returns 0 or N with certainty, so the frequency can never leave those two values (they are **absorbing**). Anywhere in between the variance $p(1 - p)/N$ is positive, so the frequency keeps moving until it hits one of them.

Now, which one? Look at the population far in the future, after fixation. Every gene copy alive then is a descendant of exactly one of the N gene copies alive today (the one whose line of descent never died out). Under pure drift nothing distinguishes today’s N copies from one another – they have the same expected number of descendants, and the same chances at every step – so each of them is equally likely to be that single common ancestor: probability $1/N$ each. Allele A is carried by $k = pN$ of today’s copies, hence

$$P(A \text{ is eventually fixed}) = pN \cdot \frac{1}{N} = p, \quad P(A \text{ is eventually lost}) = 1 - p.$$

A **new mutation** is a single copy, $p = 1/N$, so it fixes with probability $1/N$ (lecture, diploid: $1/2N$). Note that this argument never used the binomial distribution: it only needs neutrality and the fact that all copies are interchangeable.

Experimental evidence. Buri (1956) followed 107 laboratory populations of 16 *Drosophila* each, all starting at $p = 0.5$. After 19 generations most populations were fixed for one allele or the other – and, as $P(\text{fix}) = 0.5$ predicts, about half of them for each.

2.2 The substitution rate equals the mutation rate

A **substitution** is a mutation that eventually becomes fixed. If each gene copy mutates with probability μ per generation, then $N\mu$ new mutations appear in the population every generation. Each of them starts at frequency $1/N$ and therefore fixes with probability $1/N$. The expected number of mutations per generation that will eventually fix is

$$N\mu \cdot \frac{1}{N} = \mu \quad (\text{lecture, diploid: } 2N\mu \cdot \frac{1}{2N} = \mu).$$

Meaning. The population size cancels. A large population produces more mutations, but each one has a proportionally smaller chance of fixing, and the two effects exactly balance. Because the substitution rate depends only on μ , differences between species accumulate at a constant rate – the basis of the molecular clock.

2.3 How long does fixation or loss take? (take as given)

Kimura & Ohta (1969) showed that the expected number of generations until A is either fixed or lost, starting from frequency p , is

$$\bar{t}(p) = -2N [p \ln p + (1-p) \ln(1-p)] \quad (\text{lecture, diploid: } -4N[\dots]).$$

What the notebook lets you check: the curve is symmetric around $p = 1/2$ (“ A is lost” is the same event as “ a is fixed”, and the model does not care which allele we named A); its maximum is at $p = 1/2$, where $\bar{t} = 2N \ln 2 \approx 1.39 N$ (about 1390 generations for $N = 1000$); and the time is proportional to N , so doubling the population doubles the time. Conditional on it being fixed, a new mutation takes on average about $2N$ generations (lecture, diploid: $4N$), while loss of a rare allele is fast.

3 Selection

3.1 Sampling probability and expected change in frequency

Let each A copy produce $1 + s$ times as many gametes as an a copy. The gamete pool then no longer has A in proportion p : with k copies of A and $N - k$ copies of a , the A gametes make up

$$p' = \frac{k(1+s)}{k(1+s) + (N-k)} = \frac{p(1+s)}{p(1+s) + (1-p)} = \frac{p(1+s)}{1+sp}$$

of the pool (divide numerator and denominator by N to get the middle expression; expand the denominator to get the last one). The next generation is again Binomial(N, p') – this is the notebook’s the notebook function with selection. With $s = 0$, $p' = p$ and we are back to the neutral model.

The expected change per generation is

$$\Delta p = p' - p = \frac{p(1+s) - p(1+sp)}{1+sp} = \frac{p + sp - p - sp^2}{1+sp} = \frac{sp(1-p)}{1+sp} \approx sp(1-p) \quad (\text{for small } s).$$

Meaning. The factor $p(1-p)$ is largest at $p = 1/2$ and tiny when A is rare or nearly fixed. To see whether selection or drift controls a rare allele, compare the expected push, $sp(1-p)$, with the size of one random step, $\sqrt{p(1-p)/N}$ (Section 1.3). Their ratio is $s\sqrt{Np(1-p)}$, which goes to zero as $p \rightarrow 0$: while an allele is rare, drift dominates its fate no matter how beneficial it is.

3.2 Fixation probability with selection (take as given, but check the limits)

$$P(\text{fix} | p) = \frac{1 - e^{-2Nsp}}{1 - e^{-2Ns}} \quad (\text{lecture, diploid: } \frac{1 - e^{-4Nsp}}{1 - e^{-4Ns}}).$$

Here s is the advantage *per gene copy*, as in Section 3.1 (textbooks that define s for the homozygote get half these values).

- *Neutral limit.* For $s \rightarrow 0$ use $e^{-x} \approx 1 - x$ in numerator and denominator: $\frac{2Nsp}{2Ns} = p$, the result of Section 2.1.

- *What decides between drift and selection* is the product Ns (lecture, diploid: $2Ns$). $Ns \ll 1$: effectively neutral, $P(\text{fix}) \approx p$. $Ns \approx 1$: both matter; for $p = 1/2$, $P = \frac{1-e^{-1}}{1-e^{-2}} = 0.73$. $Ns \gg 1$: an allele already at intermediate frequency is almost certain to fix.
- *A new beneficial mutation*. Put $p = 1/N$ and assume $Ns \gg 1$:

$$P(\text{fix}) = \frac{1 - e^{-2s}}{1 - e^{-2Ns}} \approx 1 - e^{-2s} \approx 2s.$$

Two approximations: $e^{-2Ns} \approx 0$ because Ns is large, and $1 - e^{-2s} \approx 2s$ because s is small. The population size has disappeared: a single copy's fate is settled in its first few generations, while it is rare (Section 3.1), and at that stage the size of the rest of the population is irrelevant. With $s = 0.05$ the chance is only about 10% – most beneficial mutations are lost to drift.

4 The coalescent (backward in time)

4.1 Two lineages

Take two gene copies from today's population and follow their ancestry back one generation. Each of them was drawn from the gamete pool of generation $t - 1$, i.e. each was produced by one of the N gene copies of that generation. The first copy came from *some* parental copy; the second copy, drawn independently, came from that *same* parental copy with probability $1/N$ (one of N equally likely possibilities). If it did, the two lineages have found their common ancestor: they **coalesce**. If not (probability $1 - 1/N$), we now have two distinct copies in generation $t - 1$ and the situation is exactly as before, so the same reasoning applies in the next generation back, with the same probability $1/N$. The number of generations X_2 until they coalesce is therefore geometric:

$$P(X_2 > t) = \left(1 - \frac{1}{N}\right)^t \quad (\text{no coalescence in any of } t \text{ consecutive generations}).$$

For large N this is very close to an exponential: $\ln(1 - x) \approx -x$ for small x , so $\ln\left[\left(1 - \frac{1}{N}\right)^t\right] = t \ln\left(1 - \frac{1}{N}\right) \approx -t/N$, giving

$$P(X_2 > t) \approx e^{-t/N} \implies X_2 \sim \text{Exponential}(\text{rate } \frac{1}{N}),$$

$$E[X_2] = N, \quad \text{Var}(X_2) = N^2 \quad (\text{lecture, diploid: } 2N \text{ and } 4N^2).$$

(An exponential variable with rate λ has mean $1/\lambda$ and variance $1/\lambda^2$.) Three properties that the histogram in the notebook shows and that are easy to get wrong:

- the standard deviation equals the mean, so a single genealogy is a very noisy measurement of N ;
- the density $\frac{1}{N}e^{-t/N}$ is highest at $t = 0$: the single most probable generation for the coalescence is the very next one, even though the average wait is N generations;
- the process is **memoryless**: the chance of coalescing in the next generation is $1/N$ regardless of how many generations have already passed.

4.2 n lineages: the waiting time while there are k of them

With k lineages there are $\binom{k}{2} = k(k-1)/2$ distinct pairs, and each pair coalesces at rate $1/N$ independently of the others. The time until the *first* of several independent exponential events is again exponential, with rate equal to the *sum* of the individual rates. So the time T_k during which there are exactly k lineages is

$$T_k \sim \text{Exponential}\left(\text{rate } \frac{k(k-1)}{2N}\right), \quad E[T_k] = \frac{2N}{k(k-1)} \quad (\text{lecture, diploid: } \frac{4N}{k(k-1)}), \quad \text{Var}(T_k) = E[T_k]^2.$$

For $N = 10\,000$: $E[T_2] = 10\,000$, $E[T_5] = 1\,000$, $E[T_{10}] = 222$ – the first event happens fast when there are many lineages. Which pair coalesces is uniformly random (all pairs have the same rate), which is why the tree *shape* carries no information under neutrality. After a coalescence the process simply continues with $k - 1$ lineages. In `msprime` the sample nodes are $0, \dots, n - 1$ and the internal nodes are numbered in order of age, so node n is the first coalescence and node $2n - 2$ the root.

4.3 Time to the most recent common ancestor (tree height)

The height of the tree is the sum of the successive intervals, $t_{\text{MRCA}} = T_n + T_{n-1} + \dots + T_2$, so its expectation is $\sum_{k=2}^n 2N/(k(k-1))$. To evaluate the sum, split each term with the identity

$$\frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k} \quad \left(\text{check: } \frac{k - (k-1)}{k(k-1)} = \frac{1}{k(k-1)}\right).$$

Written out, the sum **telescopes**: every term cancels the next one, and only the first and the last survive,

$$E[t_{\text{MRCA}}] = 2N \left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n-1} - \frac{1}{n}\right) \right] = 2N \left(1 - \frac{1}{n}\right) \quad (\text{lecture, diploid: } 4N(1 - \frac{1}{n})).$$

The intervals are independent, so their variances add:

$$\text{Var}(t_{\text{MRCA}}) = \sum_{k=2}^n \left(\frac{2N}{k(k-1)} \right)^2 \quad (N = 10\,000, n = 5 : \text{mean } 16\,000, \text{sd} \approx 10\,700).$$

Meaning. For $n = 2, 5, 10, 50$ the expected height is $N, 1.6N, 1.8N, 1.96N$: it saturates at $2N$ (lecture, diploid: $4N$). Adding samples barely changes the height because the last interval alone, $E[T_2] = N$, is $\frac{1}{2(1-1/n)}$ of the total – 62.5% for $n = 5$. Most of any genealogy is the wait for the final two lineages; extra samples only add short branches near the tips.

4.4 Two subpopulations with migration

Two lineages can coalesce only if they are in the same subpopulation. Looking backward in time, a lineage moves to the other subpopulation at rate m per generation (forward in time: a fraction m of each subpopulation consists of immigrants). Two lineages that sit in different subpopulations must wait until one of them migrates; two independent events at rate m each give a combined rate $2m$, so

$$E[\text{wait for a migration}] = \frac{1}{2m} \quad (m = 10^{-6}: 500\,000 \text{ generations}).$$

Compare this with the coalescence time scale inside one subpopulation, N : the ratio $N/(1/2m) = 2Nm$ is proportional to Nm , the number of migrant copies per generation (lecture, diploid: $2Nm$). If $Nm \ll 1$ the tree consists of two clades, one per subpopulation, joined by a branch of length $\sim 1/(2m)$. If $Nm \gtrsim 1$ – about one migrant per generation – lineages move between the subpopulations faster than they coalesce, and the sample behaves as if it came from one well-mixed population of $2N$ copies, with expected height $2 \cdot 2N(1 - 1/n)$.

4.5 Bottlenecks and the effective population size

During a period of τ generations at size $N_b \ll N$, the pairwise coalescence rate is $1/N_b$ instead of $1/N$, i.e. N/N_b times higher. If τ is several times N_b , almost every lineage that enters the bottleneck coalesces inside it, so the sample's most recent common ancestor lies within the bottleneck and all older variation is erased.

More generally, a population whose size changes over time has the same amount of drift as a constant population of some **effective size** N_e . For sizes $N_1, \dots, N_t$ over t generations, the probability that two lineages have not coalesced is $\prod_i (1 - 1/N_i) \approx \exp(-\sum_i 1/N_i)$ (same approximation as in 4.1). A constant population would give $\exp(-t/N_e)$. Setting the exponents equal,

$$\frac{t}{N_e} = \sum_{i=1}^t \frac{1}{N_i} \quad \Rightarrow \quad N_e = \frac{t}{\sum_{i=1}^t 1/N_i} \quad (\text{the harmonic mean of the } N_i).$$

The harmonic mean is dominated by the smallest values (one generation at $N = 10$ among 99 at $N = 10\,000$ gives $N_e \approx 910$), which is why a short bottleneck lowers N_e far below the census size for a long time afterwards.